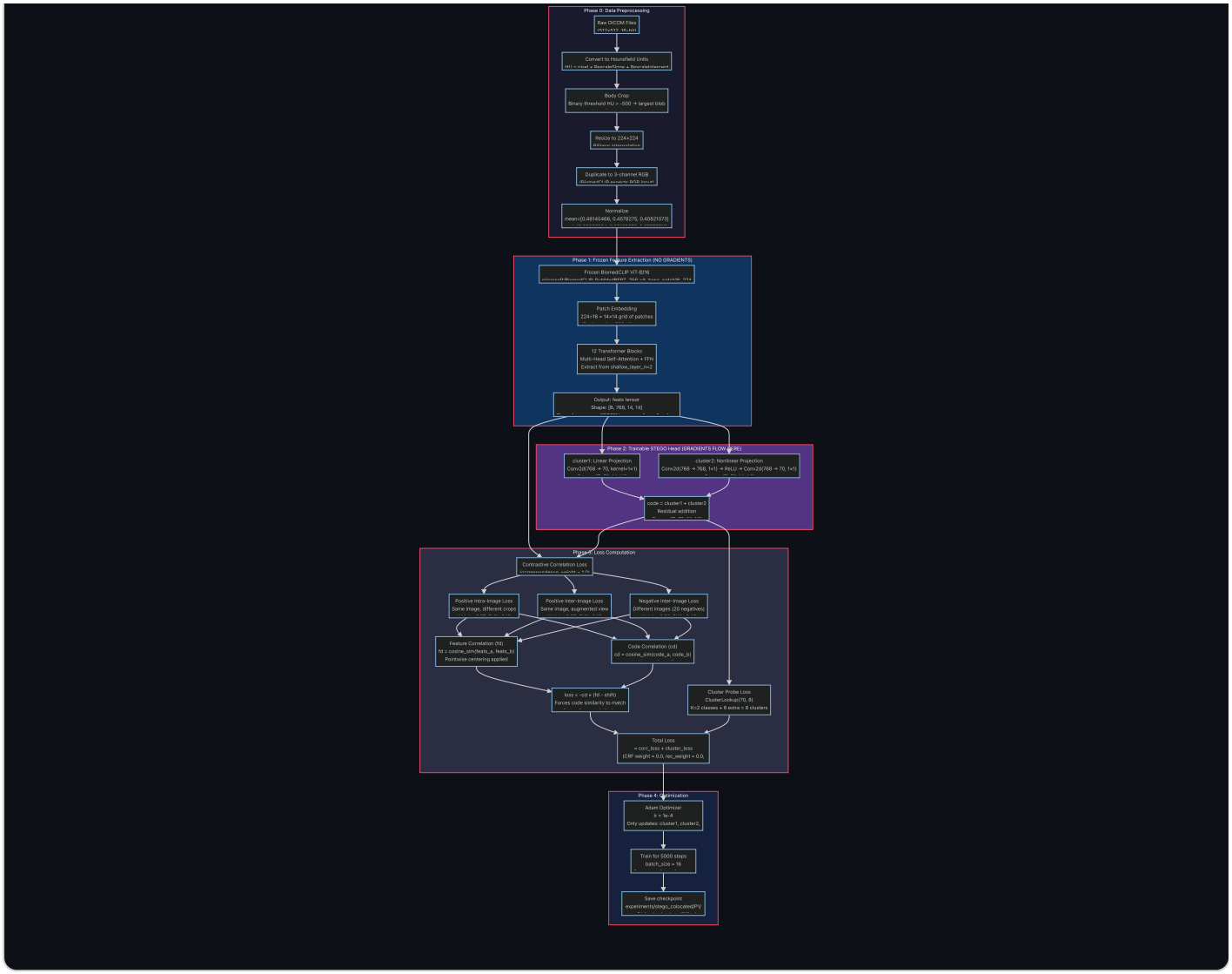
Complete Pipeline Flowchart: Trained STEGO P1 Head → 83.92% IoU

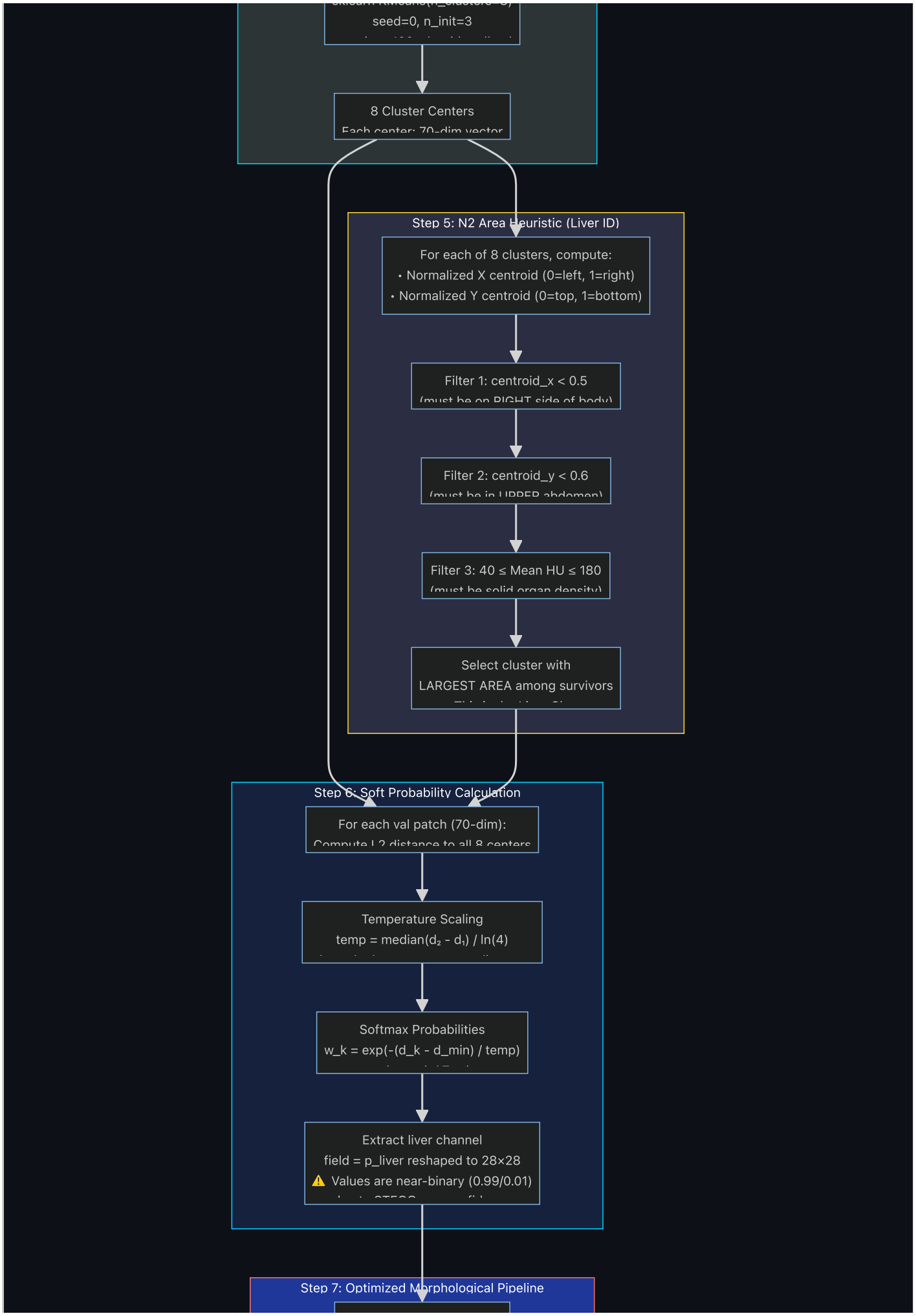
Important: > This flowchart covers **every single step** from raw DICOM pixels to the final binary mask, split into two major phases: **Training** (how the P1 Head learned its weights) and **Inference** (how the 83.92% score is produced at evaluation time).

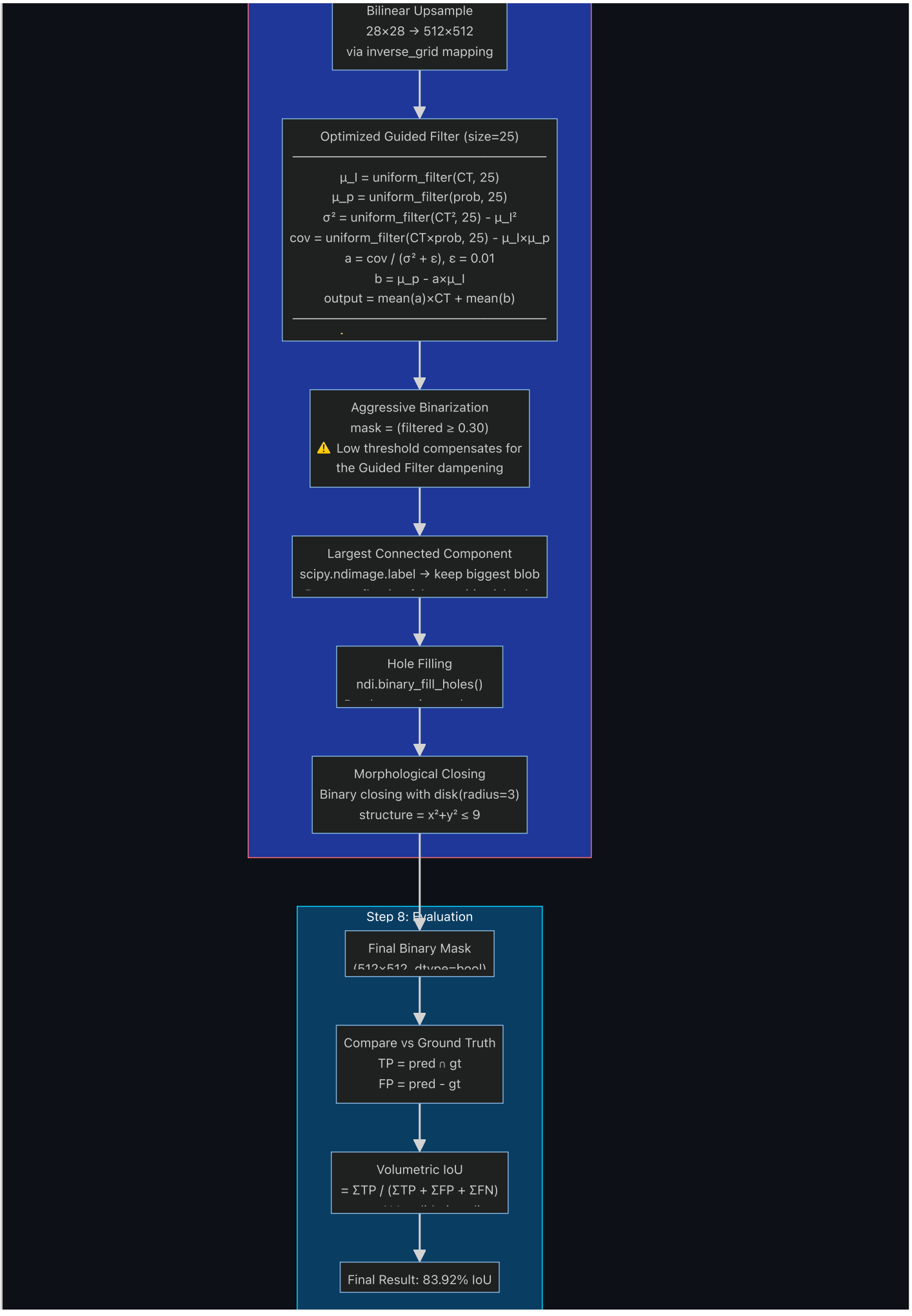
Part A: The Training Phase (How the P1 Head Was Trained)


Phase 0: Data Preprocessing
Raw DICOM Files
Convert to Hounsfield Units
Body Crop
Binary threshold HU > -500 → largest blob
Resize to 224×224
Duplicate to 3-channel RGB
Normalize
Phase 1: Frozen Feature Extraction (NO GRADIENTS)
Frozen BiomedCLIP ViT-B/16
Patch Embedding
224÷16 = 14×14 grid of patches
12 Transformer Blocks
Multi-Head Self-Attention + FFN
Extract from shallow_layer_hv2
Output: feats tensor
Phase 2: Trainable STEGO Head (GRADIENTS FLOW HERE)
cluster1: Linear Projection
Conv2d(768 → 70, kernel=1×1)
cluster2: Nonlinear Projection
Conv2d(768 → 768, 1×1) → ReLU → Conv2d(768 → 70, 1×1)
code = cluster1 + cluster2
Residual addition
Phase 3: Loss Computation
Contrastive Correlation Loss
Positive intra-image Loss
Same image, different crops
Positive inter-image Loss
Same image, augmented view
Negative inter-image Loss
Different images (20 negatives)
Feature-Correlation (fd)
fd = cosine_sim(feats_a, feats_b)
Pointwise centering applied
Code Correlation (cd)
cd = cosine_sim(code_a, code_b)
loss = −cd × (fd − shift)
Forces code similarity to match
Cluster Probe Loss
ClusterLookup(70, 8)
K=2 classes + 6 extra = 8 clusters
Total Loss
= corr_loss + cluster_loss
(CRF weight = 0.0, rec_weight = 0.0)
Phase 4: Optimization
Adam Optimizer
lr = 5e-4
Only updates: cluster1, cluster2,
Train for 6000 steps
batch_size = 16
Save checkpoint
experiments/stego_colocatedPV

Part B: The Inference Phase (How 83.92% IoU Is Produced)


Step 1: Input Preprocessing
Raw DICOM (512×512)
HU Conversion
Body Crop
threshold HU > -500
binary_opening(disk r=5)
largest connected component
Resize to 224×224
3-channel + Normalize
Step 2: Feature Extraction
Frozen BiomedCLIP ViT-B/16
Output: 14×14 × 768 features
Step 3: Trained STEGO P1 Head
Load final.pt checkpoint
cluster2: Conv2d(768→768,1×1)
→ ReLU
Output: 28×28 × 70 embeddings
⚠ These are OVERCONFIDENT
Step 4: K-Means Clustering (K=8)
Flatten all TRAIN patches
sklearn KMeans(n_clusters=8)


seed=0, n_init=3
8 Cluster Centers
Step 5: N2 Area Heuristic (Liver ID)
For each of 8 clusters, compute:
• Normalized X centroid (0=left, 1=right)
• Normalized Y centroid (0=top, 1=bottom)
Filter 1: centroid_x < 0.5
Filter 2: centroid_y < 0.6
Filter 3: 40 ≤ Mean HU ≤ 180
Select cluster with
LARGEST AREA among survivors
Step 6: Soft Probability Calculation
For each val patch (70-dim):
Temperature Scaling
temp = median(d₂ - d₁) / ln(4)
Softmax Probabilities
w_k = exp(-(d_k - d_min) / temp)
Extract liver channel
field = p_liver reshaped to 28×28
⚠ Values are near-binary (0.99/0.01)
Step 7: Optimized Morphological Pipeline


Bilinear Upsample
28×28 → 512×512
via inverse_grid mapping
Optimized Guided Filter (size=25)
μ_I = uniform_filter(CT, 25)
μ_p = uniform_filter(prob, 25)
σ² = uniform_filter(CT², 25) - μ_I²
cov = uniform_filter(CT×prob, 25) - μ_I×μ_p
a = cov / (σ² + ε), ε = 0.01
b = μ_p - a×μ_I
output = mean(a)×CT + mean(b)
Aggressive Binarization
mask = (filtered ≥ 0.30)
⚠ Low threshold compensates for the Guided Filter dampening
Largest Connected Component
scipy.ndimage.label → keep biggest blob
Hole Filling
ndi.binary_fill_holes()
Morphological Closing
Binary closing with disk(radius=3)
structure = x²+y² ≤ 9
Step 8: Evaluation
Final Binary Mask
Compare vs Ground Truth
TP = pred ∩ gt
FP = pred - gt
Volumetric IoU
= ΣTP / (ΣTP + ΣFP + ΣFN)
Final Result: 83.92% IoU

Part C: Why It's 83.92% and Not 86.72%



The Two Sources of Degradation

Factor	Frozen Baseline (86.72%)	Trained P1 Head (83.92%)	Impact
Feature Dimensionality	Raw 768-dim BiomedCLIP	Compressed to 70-dim by STEGO	Loses ~90% of feature information. Irreversible dimensionality reduction destroys subtle tissue boundaries.
Probability Distribution	Naturally fuzzy (e.g., 60%/40%)	Overconfident (e.g., 99.9%/0.1%)	The contrastive loss forces features to be maximally separable, producing cliff-like probability walls that break the Guided Filter math.
Guided Filter Behavior	Smoothly aligns gradients to CT edges	Over-expands mask into surrounding tissue	With soft inputs, `a ≈ small`, `b ≈ μ_p` → gentle edge-following. With sharp inputs, `a ≈ large` → aggressive edge-jumping, causing false positives.

File References

Component	File	Key Lines
BiomedCLIP Featurizer	`modules.py`	`class BiomedCLIPFeaturizer`
STEGO Head (cluster1 + cluster2)	`modules.py`	`make_clusterer`, `make_nonlinear_clusterer`
Contrastive Correlation Loss	`modules.py`	`class ContrastiveCorrelationLoss`
Contrastive CRF Loss	`modules.py`	`class ContrastiveCRFLoss` (weight=0.0, unused)
Training Loop & All Losses	`train_segmentation.py`	`training_step`
Training Config	`train_config.yaml`	All hyperparameters
P1 Inference + N2 Heuristic	`run_p1_optimized.py`	Complete evaluation script
Guided Filter Math	`run_p1_optimized.py`	`apply_postprocessing`